Finding a Good Research Question

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The one idea for today

A thesis is not a topic you write *about*.

It is a question you answer.

Today

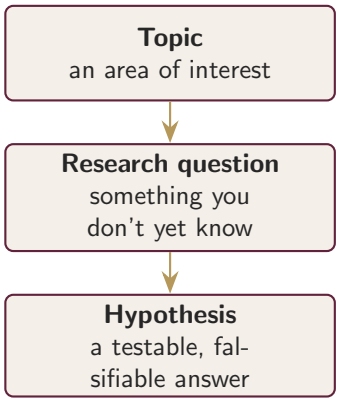
1. Topic → question → hypothesis
2. What makes a question good
3. Where good questions come from
4. Scoping to thesis size
5. Common traps & a short exercise

Coming next

2. Reading & using the literature
3. The basics of working with data

The question you leave with today is what you will take into both.

Topic $\neq$ question $\neq$ hypothesis



"Water and agriculture in the Western Cape."

"How did farms in the Western Cape adjust crop choice and irrigation during the 2015–2018 drought restrictions?"

"Farms facing tighter water allocations shifted area away from water-intensive annual crops toward permanent crops."

The one-sentence test

If you cannot fill in this sentence, you do not have a question yet:

I am studying _____,
because I want to find out _____,
so that [who] can better understand / decide _____.

- Slot 1 is your **topic**; slot 2 is your **question**; slot 3 is your **"so what"**.
- Most weak proposals have slot 1 filled in beautifully and slots 2–3 blank.

Adapted from Booth, Colomb & Williams, *The Craft of Research*.

Five tests for a good question

Answerable	Could evidence, in principle, settle it?
Feasible	With data you can get, in the time you have, with skills you have (or can learn)?
Novel	Is the answer not already known – for this place, period, or population?
Important	Does someone care about the answer?
Economic	Is there a mechanism – incentives, prices, constraints, markets?

Useful rule of thumb

You need all five, but *feasible* is the one that kills most proposals.

Novel does not mean revolutionary. A known question in a new setting, with new data, or with a better method is a perfectly good thesis.

Four kinds of questions

Type	Asks...	Agricultural economics example
Descriptive	What is? How much? Who?	How concentrated is ownership along the South African wine value chain, and how has it changed?
Causal	What is the effect of X on Y?	Did access to a contract with a packhouse raise smallholder fruit growers' incomes?
Predictive	What will happen if...?	How would a 10% rise in diesel prices affect grain farm margins in the Free State?
Evaluative	Did it work? At what cost?	Was a given extension or support programme cost-effective relative to its alternatives?

- Be honest about which type you are asking. Causal questions need a credible comparison – "impact of" is a promise about identification.
- Good descriptive work is undervalued. Many strong theses are descriptive.

The economics in agricultural economics

A question becomes an economics question when it is about choices under constraints:

- Who is deciding? (farmer, household, firm, government)
- What do they want? (profit, risk, food security)
- What constrains them? (land, water, credit, information)
- What prices or incentives change the choice?

Same topic, two disciplines

Agronomy: Which cultivar yields most under deficit irrigation?

Ag econ: Given water prices and output prices, *would farmers adopt* that cultivar – and who gains if they do?

Where good questions come from

1. **A puzzle** – something that doesn't fit what theory predicts
2. **A policy or market change** – a new tariff, a drought, a regulation, a trade shock
3. **A new dataset** – a survey round, administrative records, satellite data
4. **A disagreement** in the literature
5. **A known result, new setting** – does it hold here, now, for this group?
6. **A practitioner's problem** – from industry, a co-op, an NGO, a department

Start from a good paper

If you have a paper you like, is there something you can do with it: can you adapt the model, apply it to different data, change the question, extend it?

The “so what?” test

For your question, ask three things:

1. **Who** would want to know the answer?
2. **What would they do differently** if the answer were A rather than B?
3. **If you found nothing** (a zero effect), would that still be interesting?

× Only one answer is interesting
“Did the subsidy help?” – if no effect is found, readers assume the study was weak.

✓ Both answers are interesting
“Did the subsidy reach the intended farmers, or was it captured by larger ones?”

Scoping: narrow until it feels uncomfortable

Too broad	Food security in South Africa.
Still broad	The effect of food prices on household food security.
Thesis-sized	How did the 2022 rise in maize meal prices change food expenditure shares of low-income households, using two rounds of a national survey?
Too narrow?	Spaza-shop maize meal prices in one township in March.

A thesis-sized question names five things

1. **Who** – unit of analysis
2. **Where & when**
3. **Outcome** – what you measure
4. **Variation** – what differs, and why
5. **Evidence** – the data you'll use

Applied to the thesis-sized example

Who	Low-income households
Where/when	South Africa, before and after the 2022 price rise
Outcome	Share of spending on food
Variation	Maize meal prices over time; how much each household relies on maize meal
Evidence	Two rounds of a national household survey

Feasibility check before you decide

Data	Does it exist? Can you access it – legally, affordably, in time? Have you *seen* it, or only heard it exists?
Method	Do you know a method that answers this type of question? If not, can you learn it this year?
Time	Work backwards from the submission date. Where does data collection or cleaning fit?
Supervision	Is there someone in the department who knows this area or method?
Ethics	Human participants, farm-level data, or sensitive information mean clearance – and clearance takes weeks.

"The data exists" and "I have the data" are separated by months.

Seven common traps

1. **The topic in disguise**
   "An analysis of the citrus industry"
2. **The answer is already known**
   ...You just haven't read enough yet
3. **The recommendation in disguise**
   "Government should support..."
4. **"Impact of" with no comparison**
5. **The shopping list**
   Six objectives that are six different theses
6. **Method first**
   "I want to use machine learning on...something"
7. **Data you don't have**

Before & after

× Before	✓ After
The impact of climate change on the wine industry.	How have rising summer temperatures since 2000 been associated with changes in cultivar composition across Western Cape wine regions?
The challenges facing emerging farmers.	Which constraints – credit, land tenure, or market access – best explain differences in output among emerging livestock farmers in a given district?
Government should invest more in agricultural extension.	Do farmers who receive extension visits adopt recommended practices at higher rates, after accounting for who gets visited?

Exercise (6 minutes)

On your own (3 min)

1. Write your topic in five words.
2. Fill in the one-sentence test.
3. Underline: who, where/when, outcome, variation, evidence.

With a neighbour (3 min)

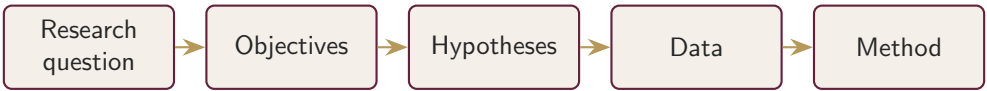
4. Swap. Which of the seven traps does theirs fall into?
5. Which of the five tests is weakest?

Don't have a topic yet?

Use one of these:

- Load-shedding and dairy farms
- Export market access for table grapes
- Land reform beneficiaries and credit
- Rooibos and climate risk

From question to proposal



- Every objective should answer part of the question. Every hypothesis should test part of an objective. Every variable should serve a hypothesis.
- If something in your proposal doesn't trace back to the question, cut it.
- Your question will change as you read! This is a good thing!

Before the next lecture

Bring to Lecture 2

1. Your question in one sentence (one-sentence test filled in).
2. A note on which type it is: descriptive, causal, predictive, evaluative.
3. Three papers you think are related (and you don't need to read them).

- Next time: we'll talk about finding papers, understanding the literature, and how to use papers to find out whether your question is new, and how to make it sharper.

Questions?

A thesis is a question you answer.

Who · where & when · outcome · variation · evidence

Further reading: Booth, Colomb & Williams, *The Craft of Research*; Thomson, *A Guide for the Young Economist*.